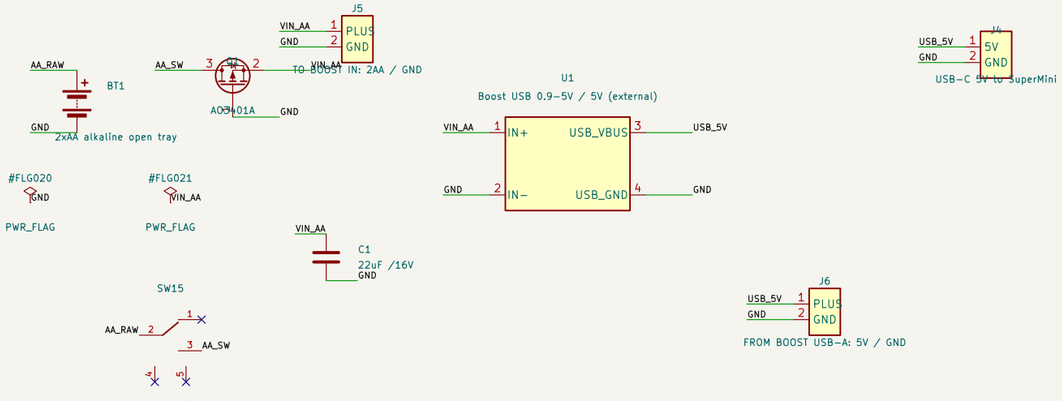
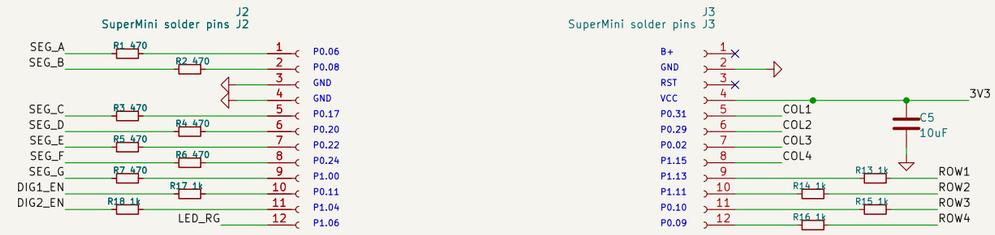
1. NGUON 2xAA: BT1 -> SW15 -> Q1 -> J5 -> external USB boost -> J6 -> J4 -> SuperMini



U1 is an EXTERNAL module; pin numbers identify logical terminals, not a PCB land pattern.
J5.1 = protected AA+ to IN+; J5.2 = GND to IN-.
J6.1 = USB-A cable VBUS +5V; J6.2 = USB-A cable GND.
J4.1 = USB-C cable VBUS +5V; J4.2 = GND.
Unplug battery USB-C cable before connecting PC.
J3.1 B+ NC; never connect primary AA to charger B+.
Use 2xAA alkaline; replace weak pair. Battery runtime must be measured.

2. CHÂN HÀN SuperMini nRF52840 (J2 = hàng chân trái, J3 = hàng chân phải; chân 1 ở đầu USB)

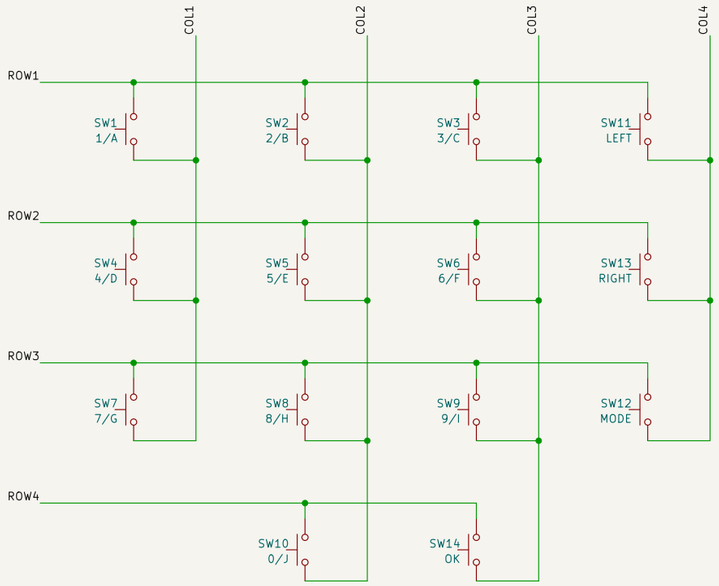
Điện trở nối tiếp GPIO; B+ J3.1 NC. J4 cấp 5V qua dây USB-C. C5 lọc VCC đầu ra.



R1-R7: dong = (VCC - VF(LED) - VOL(GPIO)) / 470 ohm.
R17/R18: 1k gate Q2/Q3; R10/R11: 100k pull-up VCC.
R13-R16: 1k noi tiep GPIO hang phim.
C1: 22uF input. C3/C4 removed; U1 external USB boost module.
C5: 10uF tai VCC. J3.1 B+ de trong.
Do VCC/GPIO, REGOUT0 va P0.13 tren SuperMini that.
Nut SMD IC3236: land 2x2mm, pitch 6.0mm.

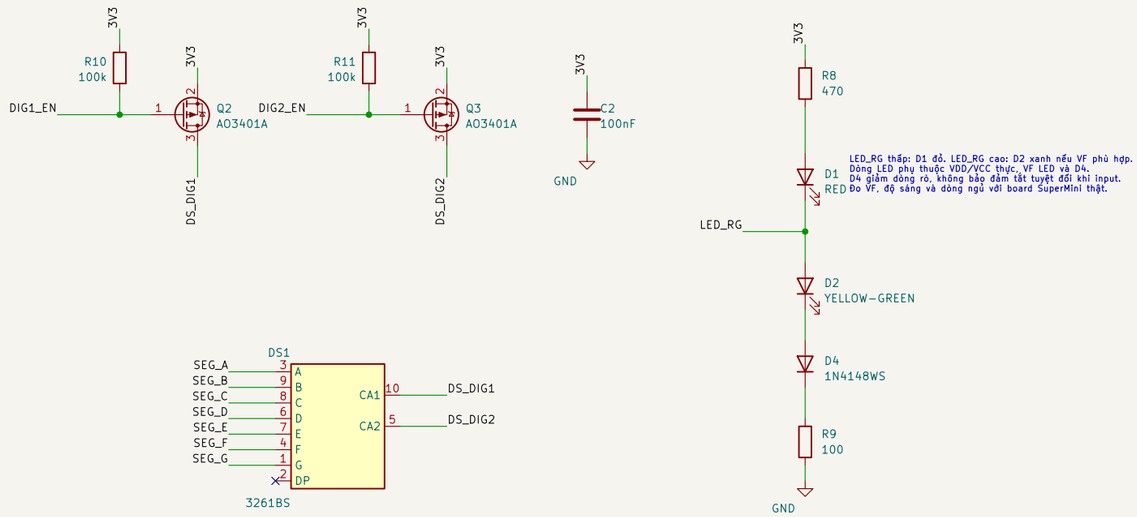
Rev J: tracks 0.25mm; clearance >=0.23mm.
GND pour + thermal gap 0.508mm; PCB 68x80mm.
SW1-SW14: IC3236, 2 SMD pads, pitch 6.0mm.

3. BÀN PHÍM MA TRẬN 4×4 (14 phím, 8 GPIO)



Hàng ROW1-4 đã qua R13-R16 (tại J3). Cột dùng pull-up nội + SENSE mức thấp để đánh thức; khi ngủ cả 4 hàng ra mức thấp.
Vị trí phím trên vỏ không cần giống ma trận điện.
Firmware: khi quét, hàng KHÔNG được chọn để INPUT (thả nổi).

4. HIỂN THỊ: LED 7 đoạn 2 số 3261BS (anode chung) + đèn đỏ/xanh 1 chân



DS1: 3261BS common anode; body 15x14x7.2mm.
Pins 1G, 2DP (NC), 3A, 4F, 5CA2, 6D, 7E, 8C, 9B, 10CA1.
Pitch 2.54mm; rows 10.16mm. Qualify generic shop sample.
For sleep: gates HIGH (digits OFF); SEG/LED_RG input; measure current.

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Size: A3	Date: 2026-10-02	**Rev: J**
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